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Filter circuit

Abstract

A filter circuit includes two ports, a first resonant circuit, and a second resonant circuit. The first resonant circuit is provided between the two ports in a circuit configuration and coupled with both of the two ports. The second resonant circuit is provided between the two ports in the circuit configuration and coupled with at least one of the two ports. Coupling of the second resonant circuit and the two ports is weaker than coupling of the first resonant circuit and the two ports.

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Patent No.	Application Date	Country	CPC
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of Japanese Priority Patent Application No. 2022-033471 filed on Mar. 4, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to a filter circuit including two resonant circuits.

2. Description of the Related Art

(3) One of electronic components used in communication apparatuses is a band-pass filter. Such a band-pass filter is desired to reduce insertion loss in the passband thereof and increase insertion loss outside the passband.

(4) US 2020/0303798 A1 describes a filter device configured by combining a band-pass filter and a band elimination filter (band stop filter). This filter device uses the band elimination filter to increase insertion loss in a frequency region higher than the passband thereof.

(5) Communication services using fifth-generation mobile communication systems (hereinafter referred to as 5G) are currently started to be provided. For 5G, the use of frequency bands of 10 GHz or higher, particularly a quasi-millimeter wave band of 10 to 30 GHz and a millimeter wave band of 30 to 300 GHz, is assumed. In these frequency bands, as the frequency bands used for fourth-generation or earlier mobile communication systems, a plurality of standards dealing with relatively close frequency bands are present. Hence, it is also desired, in a band-pass filter used in

5G, that insertion loss abruptly changes in a frequency region close to the passband of the band-pass filter.

(6) In a band-pass filter of a known art, the number of resonators constituting the band-pass filter is increased in order to obtain characteristics of abrupt change of insertion loss in a frequency region close to the passband of the band-pass filter. However, there occurs a problem that the insertion loss of the passband increases as the number of resonators increases when comparison is made with the resonators having the same Q value.

(7) Now, consider that a band-pass filter uses a band elimination filter to obtain characteristics of abrupt change of insertion loss in a frequency region close to the passband of the band-pass filter. In this case, the center frequency of the stop band of the band elimination filter need be set to a frequency close to the passband. However, there occurs a problem that the insertion loss of the passband of the band-pass filter increases.

(8) The above-described problems apply not only to a filter circuit including a band-pass filter and a band elimination filter but also to filter circuits in general each using a slave resonant circuit in order to adjust characteristics of a master resonant circuit.

SUMMARY OF THE INVENTION

(9) An object of the present invention is to provide a filter circuit that includes a master resonant circuit and a slave resonant circuit and can obtain desired characteristics while suppressing deterioration of characteristics of the filter circuit obtained by the master resonant circuit.

(10) A filter circuit of the present invention includes: two ports; a first resonant circuit including a plurality of first resonators, provided between the two ports in a circuit configuration, and coupled with both of the two ports; and a second resonant circuit including a plurality of second resonators, provided between the two ports in the circuit configuration, and coupled with at least one of the two ports. Coupling of the second resonant circuit and the two ports is weaker than coupling of the first resonant circuit and the two ports.

(11) The filter circuit of the present invention may further include: two first capacitors capacitively-coupling the first resonant circuit and the two ports; and at least one second capacitor capacitively-coupling the second resonant circuit and the two ports. Capacitance of the at least one second capacitor may be smaller than capacitance of each of the two first capacitors. Alternatively, the filter circuit of the present invention may further include at least one second capacitor capacitively-coupling the second resonant circuit and the two ports, and the first resonant circuit may be directly coupled with at least one of the two ports.

(12) In the filter circuit of the present invention, the plurality of second resonators may include a first particular resonator and a second particular resonator. The first particular resonator may be coupled with one of the two ports. The second particular resonator may be coupled with the other of the two ports. The first particular resonator and the second particular resonator may be coupled with each other.

(13) In the filter circuit of the present invention, each of the plurality of second resonators may be a resonator with open ends.

(14) In the filter circuit of the present invention, the first resonant circuit may configure a band-pass filter.

(15) In the filter circuit of the present invention, the second resonant circuit may configure a band elimination filter.

(16) In the filter circuit of the present invention, coupling of the second resonant circuit and the two ports may be weaker than coupling of the first resonant circuit and the two ports. With these configurations, according to the present invention, it is possible to provide a filter circuit that can obtain desired characteristics while suppressing deterioration of characteristics of the filter circuit obtained by a first resonant circuit.

(17) Other and further objects, features, and advantages of the present invention will appear more fully from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a circuit diagram illustrating a circuit configuration of a filter circuit according to a first embodiment of the present invention.
- (2) FIG. 2 is a perspective view illustrating an external view of a multilayered filter device including the filter circuit according to the first embodiment of the present invention.
- (3) FIG. 3 is an explanatory diagram illustrating a patterned surface of a first dielectric layer in a stack of the multilayered filter device in the first embodiment of the present invention.
- (4) FIG. 4 is an explanatory diagram illustrating a patterned surface of each of second to seventh dielectric layers in the stack of the multilayered filter device in the first embodiment of the present invention.
- (5) FIG. 5 is an explanatory diagram illustrating a patterned surface of an eighth dielectric layer in the stack of the multilayered filter device in the first embodiment of the present invention.
- (6) FIG. 6 is an explanatory diagram illustrating a patterned surface of a ninth dielectric layer in the stack of the multilayered filter device in the first embodiment of the present invention.
- (7) FIG. 7 is an explanatory diagram illustrating a patterned surface of a tenth dielectric layer in the stack of the multilayered filter device in the first embodiment of the present invention.
- (8) FIG. 8 is an explanatory diagram illustrating a patterned surface of each of eleventh to sixteenth dielectric layers in the stack of the multilayered filter device in the first embodiment of the present invention.
- (9) FIG. 9 is an explanatory diagram illustrating a terminal-formed surface of a sixteenth dielectric layer in the stack of the multilayered filter device in the first embodiment of the present invention.
- (10) FIG. 10 is a perspective view illustrating inside of the stack of the multilayered filter device in the first embodiment of the present invention.
- (11) FIG. 11 is a characteristic diagram illustrating an example of frequency response of a filter circuit according to the first embodiment of the present invention.
- (12) FIG. 12 is a characteristic diagram illustrating part of the frequency response illustrated in FIG. 11, in an enlarged manner.
- (13) FIG. 13 is a circuit diagram illustrating a circuit configuration of a filter circuit according to a second embodiment of the present invention.
- (14) FIG. 14 is an explanatory diagram illustrating a patterned surface of a first dielectric layer in a stack of a multilayered filter device in the second embodiment of the present invention.
- (15) FIG. 15 is an explanatory diagram illustrating a patterned surface of each of second to seventh dielectric layers in the stack of the multilayered filter device in the second embodiment of the present invention.
- (16) FIG. 16 is an explanatory diagram illustrating a patterned surface of an eighth dielectric layer in the stack of the multilayered filter device in the second embodiment of the present invention.
- (17) FIG. 17 is an explanatory diagram illustrating a patterned surface of a ninth dielectric layer in the stack of the multilayered filter device in the second embodiment of the present invention.
- (18) FIG. 18 is an explanatory diagram illustrating a patterned surface of a tenth dielectric layer in the stack of the multilayered filter device in the second embodiment of the present invention.
- (19) FIG. 19 is an explanatory diagram illustrating a patterned surface of each of eleventh to sixteenth dielectric layers in the stack of the multilayered filter device in the second embodiment of the present invention.
- (20) FIG. 20 is an explanatory diagram illustrating a terminal-formed surface of the sixteenth dielectric layer in the stack of the multilayered filter device in the second embodiment of the present invention.
- (21) FIG. 21 is a perspective view illustrating inside of the stack of the multilayered filter device in

the second embodiment of the present invention.

(22) FIG. 22 is a characteristic diagram illustrating an example of frequency response of a filter circuit according to the second embodiment of the present invention.

(23) FIG. 23 is a characteristic diagram illustrating part of the frequency response illustrated in FIG. 22, in an enlarged manner.

(24) FIG. 24 is a circuit diagram illustrating a circuit configuration of the filter circuit according to the second embodiment of the present invention.

(25) FIG. 25 is a circuit diagram illustrating a circuit configuration of a filter circuit according to a third embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

First Embodiment

(26) Preferred embodiments of the present invention will now be described in detail with reference to the drawings. First, reference is made to FIG. 1 to describe an overview of a configuration of a filter circuit 1 according to a first embodiment of the present invention. The filter circuit 1 includes two ports 3 and 4, a first resonant circuit 10, and a second resonant circuit 20. Each of the ports 3 and 4 is a port for input or output of a signal.

(27) In the present embodiment, the first resonant circuit 10 configures a band-pass filter, and the second resonant circuit 20 configures a band elimination filter. In the present embodiment, in particular, the first resonant circuit 10 is a master resonant circuit, and the second resonant circuit 20 is a slave resonant circuit. The filter circuit 1 as a whole functions as a band-pass filter.

(28) The first resonant circuit 10 is provided between the two ports 3 and 4 in a circuit configuration. The first resonant circuit 10 is coupled with both of the two ports 3 and 4. Note that, as used herein, the phrase “in a circuit configuration” is to describe layout in a circuit diagram, not in a physical configuration.

(29) The second resonant circuit 20 is provided between the two ports 3 and 4 in the circuit configuration. The second resonant circuit 20 is coupled with at least one of the two ports 3 and 4. In the present embodiment, in particular, the second resonant circuit 20 is coupled with both of the two ports 3 and 4. Note that, in the present embodiment, the second resonant circuit 20 is provided in parallel with the first resonant circuit 10 between the two ports 3 and 4 in the circuit configuration and is not provided between the first resonant circuit 10 and the port 3 or the port 4.

(30) The filter circuit 1 further includes two first capacitors C11 and C12 capacitive-coupling the first resonant circuit 10 and the two ports 3 and 4, respectively. The first capacitor C11 capacitive-couples the first resonant circuit 10 and the port 3. The first capacitor C12 capacitive-couples the first resonant circuit 10 and the port 4.

(31) The filter circuit 1 further includes at least one second capacitor capacitive-coupling the second resonant circuit 20 and any or both of the two ports 3 and 4. In the present embodiment, in particular, the filter circuit 1 includes two second capacitors C21 and C22 as the at least one second capacitor. The second capacitor C21 capacitive-couples the second resonant circuit 20 and the port 3. The second capacitor C22 capacitive-couples the second resonant circuit 20 and the port 4.

(32) Coupling of the second resonant circuit 20 and the two ports 3 and 4 is weaker than coupling of the first resonant circuit 10 and the two ports 3 and 4. When coupling between a resonant circuit and a port is capacitive coupling as in the present embodiment, the coupling becomes stronger as the capacitance of the capacitor capacitive-coupling the resonant circuit and the port increases. In other words, the coupling between the resonant circuit and the port becomes weaker as the capacitance decreases.

(33) In the present embodiment, the capacitance of each of the second capacitors C21 and C22 is smaller than the capacitance of each of the first capacitors C11 and C12. Consequently, each of the coupling between the second resonant circuit 20 and the port 3 and the coupling between the second resonant circuit 20 and the port 4 is weaker than each of the coupling between the first resonant circuit 10 and the port 3 and the coupling between the first resonant circuit 10 and the port

4. In an example, the capacitance of each of the second capacitors **C21** and **C22** is 0.03 pF, and the capacitance of each of the first capacitors **C11** and **C12** is 0.14 pF.

(34) Reference is now made to FIG. 1 to describe an example of a configuration of each of the first and the second resonant circuits **10** and **20**. First, the first resonant circuit **10** will be described. The first resonant circuit **10** includes a plurality of first resonators. In the present embodiment, in particular, the first resonant circuit **10** includes, as a plurality of first resonators, two first resonators **11** and **12** arranged in this order from the port-3 side in the circuit configuration. Each of the first resonators **11** and **12** is a half-wave resonator with open ends. The first resonators **11** and **12** are magnetically coupled with each other.

(35) The first resonator **11** is coupled with the port **3**. The first resonator **11** has a first end **11a** being closest to the port **3** and a second end **11b** being furthest from the port **3**. The first capacitor **C11** is provided between the first end **11a** of the first resonator **11** and the port **3** in the circuit configuration.

(36) The first resonator **12** is coupled with the port **4**. The first resonator **12** has a first end **12a** being closest to the port **4** and a second end **12b** being furthest from the port **4**. The first capacitor **C12** is provided between the first end **12a** of the first resonator **12** and the port **4** in the circuit configuration.

(37) The first resonant circuit **10** further includes a capacitor **C13** connecting the second end **11b** of the first resonator **11** and the second end **12b** of the first resonator **12**. The first resonator **11** and the first resonator **12** are magnetically coupled and also capacitive-coupled via the capacitor **C13**, with each other.

(38) Next, the second resonant circuit **20** will be described. The second resonant circuit **20** includes a plurality of second resonators. In the present embodiment, in particular, the second resonant circuit **20** includes, as a plurality of second resonators, two second resonators **21** and **22** arranged in this order from the port-3 side in the circuit configuration. Each of the second resonators **21** and **22** is a half-wave resonator with open ends. The second resonators **21** and **22** are magnetically coupled with each other. The second resonators **21** and **22** respectively corresponds to a “first particular resonator” and a “second particular resonator” in the present invention.

(39) The second resonator **21** is coupled with the port **3**. The second resonator **21** has a first end **21a** being closest to the port **3** and a second end **21b** being furthest from the port **3**. The second capacitor **C21** is provided between the first end **21a** of the second resonator **21** and the port **3** in the circuit configuration.

(40) The second resonator **22** is coupled with the port **4**. The second resonator **22** has a first end **22a** being closest to the port **4** and a second end **22b** being furthest from the port **4**. The second capacitor **C22** is provided between the first end **22a** of the second resonator **22** and the port **4** in the circuit configuration.

(41) Next, a configuration of a multilayered filter device (referred to simply as a filter device below) **2** in the present embodiment will be described. FIG. 2 is a perspective view illustrating an external view of the filter device **2**. The filter device **2** is a filter device including the filter circuit **1** illustrated in FIG. 1.

(42) The filter device **2** includes the components of the filter circuit **1** described with reference to FIG. 1 and a stack **50** for integrating the components of the filter circuit **1**. The stack **50** includes a plurality of dielectric layers stacked together and a plurality of conductor layers and a plurality of through holes formed in the plurality of dielectric layers.

(43) The stack **50** has a bottom surface **50A** and a top surface **50B** located at opposite ends of the plurality of dielectric layers in a stacking direction **T**, and four side surfaces **50C** to **50F** connecting the bottom surface **50A** and the top surface **50B**. The side surfaces **50C** and **50D** face opposite to each other, and also the side surfaces **50E** and **50F** face opposite to each other. The side surfaces **50C** to **50F** are perpendicular to the top surface **50B** and the bottom surface **50A**.

(44) Here, **X**, **Y**, and **Z** directions are defined as illustrated in FIG. 2. The **X**, **Y**, and **Z** directions are

orthogonal to one another. In the present embodiment, the Z direction is a direction parallel to the stacking direction T. The direction opposite to the X direction is $-X$ direction, the direction opposite to the Y direction is $-Y$ direction, and the direction opposite to the Z direction is $-Z$ direction.

(45) As illustrated in FIG. 2, the bottom surface **50A** is located at a $-Z$ -direction end of the stack **50**. The top surface **50B** is located at a Z -direction end of the stack **50**. The side surface **50C** is located at a $-X$ -direction end of the stack **50**. The side surface **50D** is located at an X -direction end of the stack **50**. The side surface **50E** is located at a $-Y$ -direction end of the stack **50**. The side surface **50F** is located at a Y -direction end of the stack **50**.

(46) The filter device **2** further includes terminals **511** and **661** and ground conductor layers **512** and **662**. The terminal **511** and the ground conductor layer **512** are arranged at the bottom surface **50A**. In the present embodiment, in particular, the ground conductor layer **512** covers the bottom surface **50A** almost entirely. A gap is formed between the terminal **511** and the ground conductor layer **512**.

(47) The terminal **661** and the ground conductor layer **662** are arranged at the top surface **50B**. In the present embodiment, in particular, the ground conductor layer **662** covers the top surface **50B** almost entirely. A gap is formed between the terminal **661** and the ground conductor layer **662**.

(48) The terminal **511** corresponds to the port **3**, and the terminal **661** corresponds to the port **4**. The ground conductor layers **512** and **662** are each connected to the ground.

(49) Reference is now made to FIG. 3 to FIG. 9 to describe an example of the plurality of dielectric layers and the plurality of conductor layers constituting the stack **50**. In this example, the stack **50** includes sixteen dielectric layers stacked together. The sixteen dielectric layers will be referred to below as the first to sixteenth dielectric layers in the order from bottom to top. The first to sixteenth dielectric layers are denoted by reference numerals **51** to **66**, respectively. In FIG. 3 to FIG. 9, each circle represents a through hole.

(50) FIG. 3 illustrates a patterned surface of the first dielectric layer **51**. The terminal **511** and the ground conductor layer **512** are formed on the patterned surface of the dielectric layer **51**. A particular through hole **51T1** connected to the terminal **511** is formed in the dielectric layer **51**. A plurality of through holes formed in the dielectric layer **51** excluding the particular through hole **51T1** are connected to the ground conductor layer **512**.

(51) FIG. 4 illustrates a patterned surface of each of the second to seventh dielectric layers **52** to **57**. A particular through hole **52T1** is formed in each of the dielectric layers **52** to **57**. The particular through hole **51T1** formed in the dielectric layer **51** is connected to the particular through hole **52T1** formed in the dielectric layer **52**. In the dielectric layers **52** to **57**, every vertically adjacent particular through holes **52T1** are connected to each other.

(52) FIG. 5 illustrates a patterned surface of the eighth dielectric layer **58**. Conductor layers **581**, **582** and **583** are formed on the patterned surface of the dielectric layer **58**. A particular through hole **58T1** is formed in the dielectric layer **58**. The particular through hole **52T1** formed in the dielectric layer **57** is connected to the particular through hole **58T1**.

(53) FIG. 6 illustrates a patterned surface of the ninth dielectric layer **59**. Resonator conductor layers **591**, **592**, **593**, and **594** and conductor layers **595** and **596**, and a ground conductor layer **597** are formed on the patterned surface of the dielectric layer **59**. Each of the conductor layers **591** to **596** has a first end and a second end located opposite to each other.

(54) The conductor layer **591** includes a portion extending from the first end in the Y direction and a portion extending from the second end in the X direction. The conductor layer **592** includes a portion extending from the first end in the Y direction and a portion extending from the second end in the $-X$ direction. The conductor layer **593** includes a portion extending from the first end in the $-Y$ direction and a portion extending from the second end in the X direction. The conductor layer **594** includes a portion extending from the first end in the $-Y$ direction and a portion extending from the second end in the $-X$ direction. The conductor layer **595** extends from the first end toward

the second end in the -X direction. The conductor layer **596** extends from the first end toward the second end in the X direction.

(55) A particular through hole **59T1** is formed in the dielectric layer **59**. The particular through hole **59T1** is connected to a portion of the conductor layer **596** near the first end thereof. The particular through hole **58T1** formed in the dielectric layer **58** is connected to a portion of the conductor layer **595** near the first end thereof. Part of a plurality of through holes formed in the dielectric layer **58** excluding the particular through hole **58T1** and part of a plurality of through holes formed in the dielectric layer **59** excluding the particular through hole **59T1** are connected to the ground conductor layer **597**.

(56) The first end of the conductor layer **591** is at a predetermined distance from and adjacent to a portion of the conductor layer **595** near the second end thereof. The first end of the conductor layer **592** is at a predetermined distance from and adjacent to a portion of the conductor layer **596** near the second end thereof. The second end of the conductor layer **591** and the second end of the conductor layer **592** are at a predetermined distance from each other and adjacent to each other.

(57) The first end of the conductor layer **593** is at a predetermined distance from and adjacent to a portion of the conductor layer **595** near the second end thereof. The first end of the conductor layer **594** is at a predetermined distance from and adjacent to a portion of the conductor layer **596** near the second end thereof. The second end of the conductor layer **593** and the second end of the conductor layer **594** are at a predetermined distance from and adjacent to each other. The distance between the second end of the conductor layer **593** and the second end of the conductor layer **594** is greater than the distance between the second end of the conductor layer **591** and the second end of the conductor layer **592**.

(58) FIG. 7 illustrates a patterned surface of the tenth dielectric layer **60**. Conductor layers **601** and **602** are formed on the patterned surface of the dielectric layer **60**. A particular through hole **60T1** is formed in the dielectric layer **60**. The particular through hole **59T1** formed in the dielectric layer **59** is connected to the particular through hole **60T1**.

(59) FIG. 8 illustrates a patterned surface of each of the eleventh to sixteenth dielectric layers **61** to **66**. A particular through hole **61T1** is formed in each of the dielectric layers **61** to **66**. The particular through hole **60T1** formed in the dielectric layer **60** is connected to the particular through hole **61T1** formed in the dielectric layer **61**. In the dielectric layers **61** to **66**, every vertically adjacent particular through holes **61T1** are connected to each other.

(60) FIG. 9 illustrates a terminal-formed surface being a surface opposite to the patterned surface of the sixteenth dielectric layer **66**. The terminal **661** and the ground conductor layer **662** are formed on the terminal-formed surface of the dielectric layer **66**. The particular through hole **61T1** formed in the dielectric layer **66** is connected to the terminal **661**. A plurality of through holes formed in the dielectric layer **66** excluding the particular through hole **61T1** are connected to the ground conductor layer **662**.

(61) The stack **50** illustrated in FIG. 2 is formed by stacking the first to sixteenth dielectric layers **51** to **66** such that the patterned surface of the first dielectric layer **51** also serves as the bottom surface **50A** of the stack **50** and the terminal-formed surface of the sixteenth dielectric layer **66** also serves as the top surface **50B** of the stack **50**.

(62) FIG. 10 illustrates inside of the stack **50** formed by stacking the first to sixteenth dielectric layers **51** to **66**. As illustrated in FIG. 10, the plurality of conductor layers and the plurality of through holes illustrated in FIG. 3 to FIG. 9 are stacked inside the stack **50**. The conductor layer **595** is connected to the terminal **511** via the particular through holes **51T1**, **52T1**, and **58T1**. The conductor layer **596** is connected to the terminal **661** via the particular through holes **59T1**, **60T1**, and **61T1**. The ground conductor layers **512**, **597**, and **662** are connected by the plurality of through holes excluding the particular through holes **51T1**, **52T1**, **58T1**, **59T1**, **60T1**, and **61T1**.

(63) Correspondences of the components of the filter circuit **1** illustrated in FIG. 1 with the components in the stack **50** illustrated in FIG. 4 to FIG. 8 will now be described. The first resonator

11 of the first resonant circuit **10** is formed of the resonator conductor layer **591**. The first resonator **12** of the first resonant circuit **10** is formed of the resonator conductor layer **592**. The second resonator **21** of the second resonant circuit **20** is formed of the resonator conductor layer **593**. The second resonator **22** of the second resonant circuit **20** is formed of the resonator conductor layer **594**.

(64) The first capacitor **C11** is formed of the conductor layers **581**, **591**, and **595** and the dielectric layer **58** between these conductor layers. The first capacitor **C12** is formed of the conductor layers **582**, **592**, and **596** and the dielectric layer **58** between these conductor layers. The second capacitor **C21** is formed of the conductor layers **593**, **595**, and **601** and the dielectric layer **59** between these conductor layers. The second capacitor **C22** is formed of the conductor layers **594**, **596**, and **602** and the dielectric layer **59** between these conductor layers.

(65) The capacitor **C13** is formed of the conductor layers **583**, **591**, and **592** and the dielectric layer **58** between these conductor layers.

(66) Next, the structural features of the filter device **2** in the present embodiment will now be briefly described. In the filter device **2**, the resonator conductor layers **591** to **594** are provided in a space surrounded by the ground conductor layers **512** and **662** and the plurality of through holes.

(67) In the filter device **2**, the area of each of the conductor layers **601** and **602** constituting the second capacitors **C21** and **C22** respectively is smaller than the area of each of the conductor layers **581** and **582** constituting the first capacitors **C11** and **C12** respectively.

(68) The function and effects of the filter circuit **1** according to the present embodiment will now be described. As described above, in the present embodiment, the coupling of the second resonant circuit **20** and the two ports **3** and **4** is weaker than the coupling of the first resonant circuit **10** and the two ports **3** and **4**. With this, according to the present embodiment, it is possible to incorporate the second resonant circuit **20** into the filter circuit **1** while suppressing the effect of the second resonant circuit **20**.

(69) In the present embodiment, in particular, the first resonant circuit **10** configures a band-pass filter, and the second resonant circuit **20** configures a band elimination filter. The effect of the second resonant circuit **20** is concretely to increase, in frequency response of the insertion loss of the filter circuit **1** (frequency response of the insertion loss of the band-pass filter), the insertion loss of the frequency region close to the center frequency of the stop band of the band elimination filter configured by the second resonant circuit **20**. Hence, according to the present embodiment, it is possible to incorporate the second resonant circuit **20** into the filter circuit **1** while reducing the insertion loss in the frequency region to a required amount. Hence, according to the present embodiment, the center frequency of the stop band of the band elimination filter configured by the second resonant circuit **20** is adjusted to a frequency close to the passband of the band-pass filter configured by the first resonant circuit **10**, to thereby be able to obtain characteristics of abrupt change of the insertion loss in the frequency region close to the passband of the filter circuit **1**, while suppressing an increase of the insertion loss of the passband of the filter circuit **1**.

(70) Note that, also by increasing the number of resonators configuring the band-pass filter, it is possible to obtain characteristics of abrupt change of insertion loss in a frequency region close to the passband of the band-pass filter. However, there occurs a problem that the insertion loss of the passband increases as the number of resonators increases when comparison is made with the resonators having the same Q value.

(71) In contrast to this, in the present embodiment, the number of resonators included in the first resonant circuit **10** is only two. According to the present embodiment, it is possible to obtain characteristics of abrupt change of insertion loss in a frequency region close to the passband without increasing the number of resonators constituting the band-pass filter. With this, according to the present embodiment, it is possible to suppress an increase of insertion loss of the passband. According to the present embodiment, it is possible to miniaturize the filter circuit **1** and the filter device **2**.

(72) The center frequency of the stop band of the band elimination filter configured by the second resonant circuit **20** may be present in a lower frequency region from the passband of the band-pass filter configured by the first resonant circuit **10** or may be present in a higher frequency region from the passband.

(73) Next, an example of frequency response of the filter circuit **1** according to the present embodiment will be described. FIG. **11** is a characteristic diagram illustrating an example of frequency response of the filter circuit **1**. FIG. **12** is a characteristic diagram illustrating part of the frequency response illustrated in FIG. **11**, concretely a frequency region near the passband, in an enlarged manner. In each of FIG. **11** and FIG. **12**, the horizontal axis represents frequency, and the vertical axis represents attenuation. In each of FIG. **11** and FIG. **12**, the curved line to which the reference numeral **91** is added represents insertion loss, and the curved line to which the reference numeral **92** is added represents reflection loss.

(74) In the example illustrated in FIG. **11** and FIG. **12**, the center frequency of the stop band of the band elimination filter configured by the second resonant circuit **20** is present in a lower frequency region from the passband of the band-pass filter configured by the first resonant circuit **10**. As illustrated in FIG. **11** and FIG. **12**, according to the present embodiment, it is possible to obtain characteristics of abrupt change of insertion loss (attenuation) in a frequency region close to the passband. The amount of insertion loss in the passband (absolute value of attenuation) is a sufficiently small value.

Second Embodiment

(75) A second embodiment of the present invention will now be described. First, reference is made to FIG. **13** to briefly describe a filter circuit **1** according to the present embodiment in terms of differences compared to the filter circuit of the first embodiment. FIG. **13** is a circuit diagram illustrating a circuit configuration of the filter circuit **1** according to the present embodiment.

(76) In the present embodiment, each of the first resonators **11** and **12** of the first resonant circuit **10** is a quarter-wave resonator with one end being short-circuited and the other end being open. The second end **11b** of the first resonator **11** and the second end **12b** of the first resonator **12** are each connected to the ground. In FIG. **13**, the reference numeral **L11** represents an inductance component of a line connecting the first resonators **11** and **12** and the ground.

(77) Reference is now made to FIG. **14** to FIG. **20** to describe the filter device **2** according to the present embodiment. The configuration of the filter device **2** according to the present embodiment is the same as the configuration of the filter device **2** of the first embodiment excluding a plurality of dielectric layers constituting the stack **50**. In the present embodiment, the stack **50** includes sixteen dielectric layers **71** to **86** stacked together instead of the dielectric layers **51** to **66** of the first embodiment. In the following, the sixteen dielectric layers **71** to **86** will be referred to as the first to sixteenth dielectric layers **71** to **86** in the order from bottom to top. In FIG. **14** to FIG. **20**, each circle represents a through hole.

(78) FIG. **14** illustrates a patterned surface of the first dielectric layer **71**. The terminal **511** and the ground conductor layer **512** are formed on the patterned surface of the dielectric layer **71**. A particular through hole **71T1** connected to the terminal **511** is formed in the dielectric layer **71**. A plurality of through holes formed in the dielectric layer **71** excluding the particular through hole **71T1** are connected to the ground conductor layer **512**. The plurality of through holes connected to the ground conductor layer **512** include particular through holes **71T2** and **71T3**.

(79) FIG. **15** illustrates a patterned surface of each of the second to seventh dielectric layers **72** to **77**. Particular through holes **72T1**, **72T2**, and **72T3** are formed in each of the dielectric layers **72** to **77**. The particular through holes **71T1** to **71T3** formed in the dielectric layer **71** are connected respectively to the particular through holes **72T1** to **72T3** formed in the dielectric layer **72**. In the dielectric layers **72** to **77**, every vertically adjacent through holes denoted by the same reference signs are connected to each other.

(80) FIG. **16** illustrates a patterned surface of the eighth dielectric layer **78**. Conductor layers **781**

and **782** are formed on the patterned surface of the dielectric layer **78**. Particular through holes **78T1**, **78T2**, and **78T3** are formed in the dielectric layer **78**. The particular through holes **72T1** to **72T3** formed in the dielectric layer **77** are connected to the particular through holes **78T1** to **78T3**, respectively.

(81) FIG. **17** illustrates a patterned surface of the ninth dielectric layer **79**. Resonator conductor layers **791**, **792**, **793**, and **794** and conductor layers **795** and **796**, and a ground conductor layer **797** are formed on the patterned surface of the dielectric layer **79**. Each of the conductor layers **791** to **796** has a first end and a second end located opposite to each other.

(82) The conductor layer **791** and **795** each extend from the first end toward the second end in the $-X$ direction. The conductor layers **792** and **796** each extend from the first end toward the second end in the X direction. The conductor layers **793** and **794** each extend from the first end toward the second end in the $-Y$ direction.

(83) Particular through holes **79T1**, **79T2**, and **79T3** are formed in the dielectric layer **79**. The particular through hole **79T1** is connected to a portion of the conductor layer **796** near the first end thereof. The particular through hole **78T1** formed in the dielectric layer **78** is connected to a portion of the conductor layer **795** near the first end thereof. The particular through holes **78T2** and **78T3** formed in the dielectric layer **78** and the particular through holes **79T2** and **79T3** are connected to the ground conductor layer **797**.

(84) The portion near the first end of the conductor layer **791** is at a predetermined distance from and adjacent to a portion of the conductor layer **795** near the second end thereof. The portion near the first end of the conductor layer **792** is at a predetermined distance from and adjacent to a portion of the conductor layer **796** near the second end thereof. The second ends of the conductor layers **791** and **792** are connected to the ground conductor layer **797**. In FIG. **17**, the boundary between each of the conductor layers **791** and **792** and the ground conductor layer **797** is indicated by a dotted line.

(85) The first end of the conductor layer **793** is at a predetermined distance from and adjacent to a portion of the conductor layer **795** near the second end thereof. The first end of the conductor layer **794** is at a predetermined distance from and adjacent to a portion of the conductor layer **796** near the second end thereof.

(86) FIG. **18** illustrates a patterned surface of the tenth dielectric layer **80**. Conductor layers **801** and **802** are formed on the patterned surface of the dielectric layer **80**. Particular through holes **80T1**, **80T2**, and **80T3** are formed in the dielectric layer **80**. The particular through holes **79T1** to **79T3** formed in the dielectric layer **79** are connected to the particular through holes **80T1** to **80T3**, respectively.

(87) FIG. **19** illustrates a patterned surface of each of the eleventh to sixteenth dielectric layers **81** to **86**. Particular through holes **81T1**, **81T2**, and **81T3** are formed in each of the dielectric layers **81** to **86**. The particular through holes **80T1** to **80T3** formed in the dielectric layer **80** are connected respectively to the particular through holes **81T1** to **81T3** formed in the dielectric layer **81**. In the dielectric layers **81** to **86**, every vertically adjacent through holes denoted by the same reference signs are connected to each other.

(88) FIG. **20** illustrates a terminal-formed surface being a surface opposite to the patterned surface of the sixteenth dielectric layer **86**. A terminal **861** and a ground conductor layer **862** are formed on the terminal-formed surface of the dielectric layer **86**. The particular through holes **81T1** formed in the dielectric layer **86** is connected to the terminal **861**. A plurality of through holes formed in the dielectric layer **86** including the particular through holes **81T2** and **81T3** formed in the dielectric layer **86** (excluding the particular through hole **81T1**) are connected to the ground conductor layer **862**.

(89) The stack **50** in the present embodiment is formed by stacking the first to sixteenth dielectric layers **71** to **86** such that the patterned surface of the first dielectric layer **71** also serves as the bottom surface **50A** of the stack **50** and the terminal-formed surface of the sixteenth dielectric layer

86 also serves as the top surface **50B** of the stack **50**.

(90) FIG. **21** illustrates inside of the stack **50** formed by stacking the first to sixteenth dielectric layers **71** to **86**. As illustrated in FIG. **21**, the plurality of conductor layers and the plurality of through holes illustrated in FIG. **14** to FIG. **20** are stacked inside the stack **50**. The conductor layer **795** is connected to the terminal **511** via the particular through holes **71T1**, **72T1**, and **78T1**. The conductor layer **796** is connected to the terminal **661** via the particular through holes **79T1**, **80T1**, and **81T1**. The ground conductor layers **512**, **662**, and **797** are connected to each other via the plurality of through holes excluding the particular through holes **51T1**, **52T1**, **58T1**, **59T1**, **60T1**, and **61T1**. In particular, the ground conductor layer **797** is connected to the ground conductor layer **512** via the particular through holes **71T2**, **71T3**, **72T2**, **72T3**, **78T2**, and **78T3** and is connected to the ground conductor layer **662** via the particular through holes **79T2**, **79T3**, **80T2**, **80T3**, **81T2**, and **81T3**.

(91) Correspondences of the components of the filter circuit **1** illustrated in FIG. **13** with the components in the stack **50** illustrated in FIG. **15** to FIG. **19** will now be described. The first resonator **11** of the first resonant circuit **10** is formed of the resonator conductor layer **791**. The first resonator **12** of the first resonant circuit **10** is formed of the resonator conductor layer **792**. The second resonator **21** of the second resonant circuit **20** is formed of the resonator conductor layer **793**. The second resonator **22** of the second resonant circuit **20** is formed of the resonator conductor layer **794**.

(92) The first capacitor **C11** is formed of the conductor layers **781**, **791**, and **795** and the dielectric layer **78** between these conductor layers. The first capacitor **C12** is formed of the conductor layers **782**, **792**, and **796** and the dielectric layer **78** between these conductor layers. The second capacitor **C21** is formed of the conductor layers **793**, **795**, and **801** and the dielectric layer **79** between these conductor layers. The second capacitor **C22** is formed of the conductor layers **794**, **796**, and **802** and the dielectric layer **79** between these conductor layers.

(93) Next, the structural features of the filter device **2** in the present embodiment will now be briefly described. In the filter device **2**, the resonator conductor layers **791** to **794** are provided in a space surrounded by the ground conductor layers **512** and **662** and the plurality of through holes.

(94) In the filter device **2**, the area of each of the conductor layers **801** and **802** constituting the second capacitors **C21** and **C22** respectively is smaller than the area of each of the conductor layers **781** and **782** constituting the first capacitors **C11** and **C12** respectively.

(95) The particular through holes **71T2**, **71T3**, **72T2**, **72T3**, **78T2**, **78T3**, **79T2**, **79T3**, **80T2**, **80T3**, **81T2**, and **81T3** are electrically connected to the ground conductor layers **512**, **662**, and **797**. The ground conductor layers **512**, **662**, and **797** are electrically connected to the ground. In the following, the particular through holes **71T2**, **71T3**, **72T2**, **72T3**, **78T2**, **78T3**, **79T2**, **79T3**, **80T2**, **80T3**, **81T2**, and **81T3** are referred to as a plurality of particular through holes connected to the ground.

(96) The plurality of particular through holes connected to the ground include two through holes arranged in a direction orthogonal to the stacking direction **T**. The two through holes are concretely a pair of particular through holes **71T2** and **71T3**, a pair of particular through holes **72T2** and **72T3**, a pair of particular through holes **78T2** and **78T3**, a pair of particular through holes **79T2** and **79T3**, a pair of particular through holes **80T2** and **80T3**, or a pair of particular through holes **81T2** and **81T3**. The two particular through holes included in these pairs are arranged in a direction orthogonal to at least one of the direction in which the resonator conductor layer **791** extends and the direction in which the resonator conductor layer **792** extends to be described later. In the present embodiment, the two particular through holes included in these pairs are arranged in a direction parallel to the **Y** direction.

(97) The resonator conductor layer **791** extends in a first direction becoming away from the plurality of particular through holes connected to the ground. The resonator conductor layer **791** extends in a second direction becoming away from the plurality of particular through holes

connected to the ground. In the present embodiment, in particular, the resonator conductor layers **791** and **792** are each electrically connected to the plurality of particular through holes connected to the ground.

(98) The first and second directions are each a direction orthogonal to the stacking direction T. In the present embodiment, in particular, the first direction is the X direction, and the second direction is the -X direction. Hence, the first direction and the second direction are directions opposite to each other.

(99) The resonator conductor layers **793** and **794** each include a narrow portion and two wide portions located at both sides of the narrow portion. The second resonators **21** and **22** formed of the resonator conductor layers **793** and **794** are each a stepped impedance resonator.

(100) The particular function and effects of the filter device **2** in the present embodiment will now be described. In the present embodiment, the resonator conductor layers **791** and **792** each extend in the direction becoming away from the plurality of particular through holes connected to the ground as described above. Hence, in the present embodiment, when the resonator conductor layers **791** and **792** or the plurality of particular through holes connected to the resonator conductor layers **791** and **792** are shifted in a direction parallel to the X direction due to manufacturing variations, one of the resonator conductor layers **791** and **792** is longer while the other is shorter. With this, according to the present embodiment, it is possible to cancel characteristic change of resonators attributable to change in length of the resonator conductor layers. Consequently, according to the present embodiment, it is possible to suppress characteristic change of the first resonant circuit **10**, i.e., the band-pass filter, due to manufacturing variations.

(101) In the present embodiment, the plurality of particular through holes connected to the ground include the two through holes arranged in a direction orthogonal to the stacking direction T and also orthogonal to at least one of the direction in which the resonator conductor layer **791** extends and the direction in which the resonator conductor layer **792** extends. In the present embodiment, in particular, the two through holes are arranged in a direction orthogonal to both the direction in which the resonator conductor layer **791** extends and the direction in which the resonator conductor layer **792** extends. Hence, when the resonator conductor layers **791** and **792** or the plurality of particular through holes connected to the resonator conductor layers **791** and **792** are shifted in a direction parallel to the Y direction, the resonator conductor layers **791** and **792** change little in length. Also with this, according to the present embodiment, it is possible to suppress characteristic change of the first resonant circuit **10**, i.e., the band-pass filter, due to manufacturing variations.

(102) Now, the foregoing effects of the present embodiment will be described with reference to results of a simulation. In the simulation, a model of an example and a model of a comparative example were used. The model of the example and the model of the comparative example are both models of a band-pass filter including a ground conductor layer and two resonator conductor layers extending from the ground conductor layer.

(103) In the model of the example, as in the filter device **2** of the present embodiment, the two resonator conductor layers are arranged so as to sandwich the ground conductor layer and extend in directions opposite to each other. In the model of the comparative example, the two resonator conductor layers extend from the ground conductor layer in the same direction. Note that, in the simulation, the longitudinal direction of the resonator conductor layers (directions parallel to the extending directions) were the same between the model of the example and the model of the comparative example. In the simulation, the length of each of the two resonator conductor layers of the model of the example was 700 μm , and the length of each of the two resonator conductor layers of the model of the comparative example was 855 μm .

(104) In the simulation, obtained were the shift amount of the low-frequency cutoff, which is the lower limit of the passband, and the shift amount of the high-frequency cutoff, which is the upper limit of the passband, when the two resonator conductor layers were shifted by 15 μm in the longitudinal direction of the resonator conductor layers. In the model of the example, when the two

resonator conductor layers are shifted by $15\text{ }\mu\text{m}$ in the longitudinal direction of the resonator conductor layers, one of the two resonator conductor layers becomes shorter by $15\text{ }\mu\text{m}$ while the other becomes longer by $15\text{ }\mu\text{m}$. In the model of the comparative example, when the two resonator conductor layers are shifted by $15\text{ }\mu\text{m}$ in the longitudinal direction of the resonator conductor layers, both of the two resonator conductor layers become shorter by $15\text{ }\mu\text{m}$ or become longer by $15\text{ }\mu\text{m}$. In the simulation, the two resonator conductor layers were shifted so that both of the two resonator conductor layers would become longer.

(105) In a case where the two resonator conductor layers were shifted as described above, the shift amount of the low-frequency cutoff was 0.80% while the shift amount of the high-frequency cutoff was 1.25% in the model of the comparative example. In the model of the example, the shift amount of the low-frequency cutoff was 0.11% while the shift amount of the high-frequency cutoff was 0.11%. As understood from the results of the simulation, according to the present embodiment, it is possible to suppress change in low-frequency cutoff and high-frequency cutoff due to manufacturing variations.

(106) Next, an example of frequency response of the filter circuit **1** according to the present embodiment will be described. FIG. **22** is a characteristic diagram illustrating an example of frequency response of the filter circuit **1**. FIG. **23** is a characteristic diagram illustrating part of the frequency response illustrated in FIG. **22**, concretely a frequency region near the passband, in an enlarged manner. In each of FIG. **22** and FIG. **23**, the horizontal axis represents frequency, and the vertical axis represents attenuation. In each of FIG. **22** and FIG. **23**, the curved line to which the reference numeral **93** is added represents insertion loss, and the curved line to which the reference numeral **94** is added represents reflection loss.

(107) In the example illustrated in FIG. **22** and FIG. **23**, the center frequency of the stop band of the band elimination filter configured by the second resonant circuit **20** is present in a lower frequency region from the passband of the band-pass filter configured by the first resonant circuit **10**. As illustrated in FIG. **22** and FIG. **23**, according to the present embodiment, it is possible to obtain characteristics of abrupt change of insertion loss (attenuation) in a frequency region close to the passband of the band-pass filter. The amount of insertion loss in the passband (absolute value of attenuation) is a sufficiently small value.

(108) The other configuration, function, and effects of the present embodiment are the same as those of the first embodiment.

Third Embodiment

(109) A third embodiment of the present invention will now be described with reference to FIG. **24**. FIG. **24** is a circuit diagram illustrating a circuit configuration of the filter circuit **1** according to the present embodiment.

(110) The configuration of the filter circuit **1** according to the present embodiment is different from that of the first embodiment in terms of the following respects. In the present embodiment, the first capacitors **C11** and **C12** of the first embodiment are not provided. Hence, the first resonant circuit **10** is directly coupled with the ports **3** and **4**. Concretely, the first end **11a** of the first resonator **11** of the first resonant circuit **10** is directly coupled with the port **3**, and the first end **12a** of the first resonator **12** of the first resonant circuit **10** is directly coupled with the port **4**.

(111) When coupling between a resonant circuit and a port is capacitive coupling as described in the first embodiment, the coupling becomes stronger as the capacitance of the capacitor capacitive-coupling the resonant circuit and the port increases. Here, when the resonant circuit and the port are directly coupled with each other, substantially the same applies to a high-frequency region as that of a case of capacitive coupling using infinite capacitance. Accordingly, in the present embodiment, each of the coupling between the first resonant circuit **10** and the port **3** and the coupling between the first resonant circuit **10** and the port **4** is stronger than corresponding one of the first embodiment. Coupling of the first resonant circuit **10** and the two ports **3** and **4** is stronger than coupling of the second resonant circuit **20** and one of the ports **3** and **4**.

(112) Note that the configuration of the first resonant circuit **10** may be the same as that of the second embodiment. The other configuration, function, and effects of the present embodiment are similar to those of the first or second embodiment.

Fourth Embodiment

(113) Reference is now made to FIG. **25** to describe a fourth embodiment of the present invention. FIG. **25** is a circuit diagram illustrating a circuit configuration of a filter circuit **1** according to the present embodiment.

(114) The configuration of the filter circuit **1** according to the present embodiment is different from that of the first embodiment in terms of the following respects. In the present embodiment, the second resonators **21** and **22** of the second resonant circuit **20** are not magnetically coupled with each other. This configuration can be obtained, for example, by increasing the distance between the two resonator conductor layers constituting the second resonators **21** and **22**.

(115) Note that, in the present embodiment, one of the second resonators **21** and **22** need not necessarily be provided. For example, when only the second resonator **21** is provided, the second resonant circuit **20** is coupled only with the port **3**. In contrast, when only the second resonator **22** is provided, the second resonant circuit **20** is coupled only with the port **4**.

(116) The configuration of the first resonant circuit **10** may be the same as that of the second embodiment. The first resonant circuit **10** may be directly coupled with each of the ports **3** and **4** as in the third embodiment. The other configuration, function, and effects of the present embodiment are similar to any of the first to third embodiments.

(117) Note that the present invention is not limited to the foregoing embodiments, and various modifications may be made thereto. For example, the number and the configuration of each of the first and second resonators are not limited to those illustrated in the foregoing embodiments, and can be freely chosen as far as the requirements of the appended claims are met. Each of the number of first resonators and the number of second resonators may be three or more.

(118) The first resonant circuit **10** is not limited to a band-pass filter and may be a circuit configuring another filter such as a low-pass filter or a high-pass filter.

(119) Obviously, many modifications and variations of the present invention are possible in the light of the above teachings. Thus, it is to be understood that, within the scope of the appended claims and equivalents thereof, the present invention may be practiced in other embodiments than the foregoing most preferable embodiments.

Claims

1. A filter circuit comprising: two ports; a first resonant circuit including a plurality of first resonators, provided between the two ports in a circuit configuration, and coupled with both of the two ports; a second resonant circuit including a plurality of second resonators, provided between the two ports in the circuit configuration, and coupled with at least one of the two ports; and a stack including a plurality of dielectric layers stacked together, and a plurality of conductor layers and a plurality of through holes formed on the plurality of dielectric layers, the stack being for integrating the two ports, the first resonant circuit, and the second resonant circuit, wherein components of the first resonant circuit and components of the second resonant circuit are configured using the plurality of conductor layers, and coupling of the second resonant circuit and the two ports is weaker than coupling of the first resonant circuit and the two ports.
2. The filter circuit according to claim 1, further comprising: two first capacitors capacitive-coupling the first resonant circuit and the two ports; and at least one second capacitor capacitive-coupling the second resonant circuit and the two ports, wherein capacitance of the at least one second capacitor is smaller than capacitance of each of the two first capacitors.
3. The filter circuit according to claim 1, further comprising at least one second capacitor capacitive-coupling the second resonant circuit and the two ports, wherein the first resonant circuit

is directly coupled with at least one of the two ports.

4. The filter circuit according to claim 1, wherein the plurality of second resonators include a first particular resonator and a second particular resonator, the first particular resonator is coupled with one of the two ports, and the second particular resonator is coupled with the other of the two ports.
 5. The filter circuit according to claim 4, wherein the first particular resonator and the second particular resonator are coupled with each other.
 6. The filter circuit according to claim 1, wherein each of the plurality of second resonators is a resonator with open ends.
 7. The filter circuit according to claim 1, wherein the first resonant circuit configures a band-pass filter.
 8. The filter circuit according to claim 1, wherein the second resonant circuit configures a band elimination filter.
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